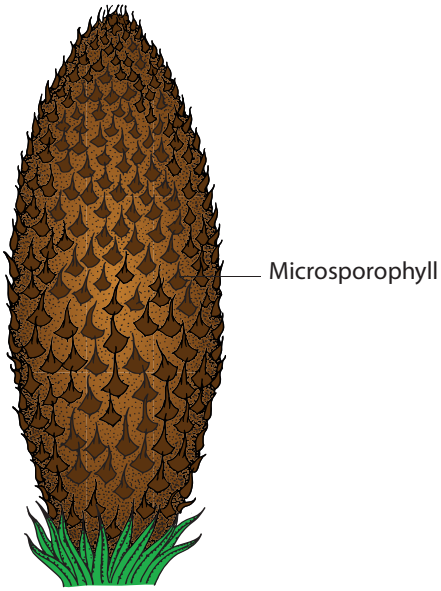
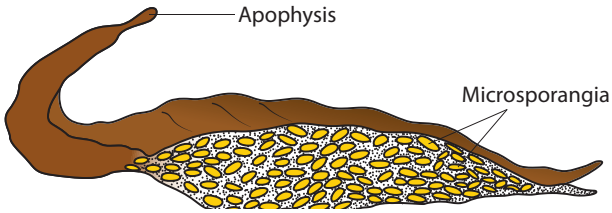
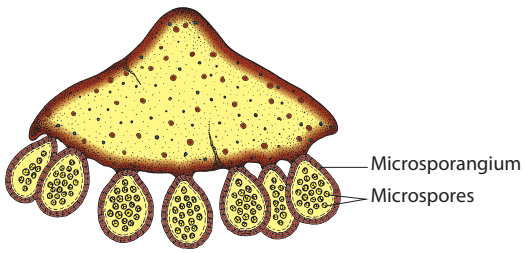
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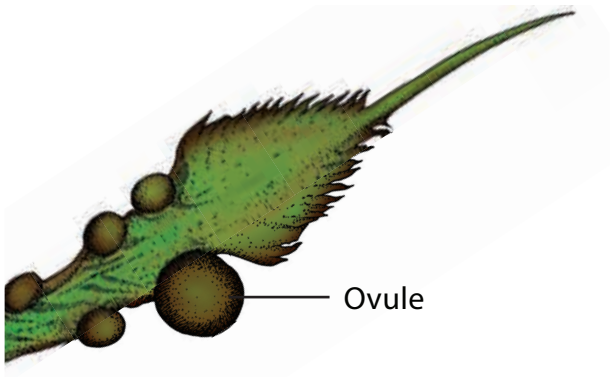
a) Male cone entire



b) A microsporophyll enlarged



c) T.S. of Microsporophyll



d) Megasporophyll

Figure 2.45: Reproduction in *Cycas*

Microsporophylls

Microsporophylls are flat, leaf-like and woody structures with narrow base and expanded upper portion. The upper expanded portion becomes pointed and is called apophysis. The narrow base is attached to the cone axis. Each microsporophyll contains thousands of microsporangia in groups called sori on abaxial (lower) surface. Development of sporangium is of Eusporangiate type. The spore mother cell undergoes meiosis to produce halpoid microspores. Each Microsporangium bears large number of microspores or pollen grains. Each sporangium is provided with a radial line of dehiscence, which helps in the dispersal of spores. Each microspore (Pollen grain) is a rounded, unicellular and uninucleate structure surrounded by outer thick exine and an inner thin intine. The microspore represents the male gametophyte.

Megasporophylls

The megasporophylls of *Cycas* are not organised into cones. They occur in close spirals around the tip of the stem of female plant. The megasporophylls are flat and measuring 15-30 cm in length. Each megasporophyll is differentiated into a basal stalk and an upper leaf like portion. The ovules are attached to the lateral side of the sporophyll. The ovules contain megaspore and it represent the female gametophyte.

Structure of Ovule

Cycas produces the largest ovule of the plant kingdom. The ovules are orthotropous, unitegmic and possess a short stalk. The single integument is very thick and covers the ovule leaving a small opening called *micropyle*. The integument